



# Institute of IT, University of Dhaka

## Training Program on AI-Assisted Data Analysis



**Training Objective:** Learn how modern data analyst and data scientists leverage AI to analysis data for their research and official activities.

**Training Description:** The training consists of 8 lecture classes, each 2 hours long and a final exam with project showcasing. Sessions are designed to gradually build students' skills through demonstrations, guided exercises, and hands-on practice in Jupyter Notebook with AI Assistant support. Each class introduces a focused topic, from data collection and cleaning to exploratory analysis, applied AI, and research reporting, ensuring students develop a complete, practical workflow for AI-driven data analysis. The final are dedicated to a mini-project where participants apply all learned techniques to solve a real research problem and present their findings.

| 8               | 16            | 4 hrs          | 1            |
|-----------------|---------------|----------------|--------------|
| LECTURE CLASSES | CONTACT HOURS | EXAM & PROJECT | MINI PROJECT |

| Lecture | Module Name                                      | Lecture Contents                                                                                                                                                                                             | Duration |
|---------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1       | Introduction to AI & Data Science                | <ul style="list-style-type: none"><li>Overview of AI, data in research, types of data, Real-life examples; data formats; discussion on datasets from different domains</li></ul>                             | 2 hours  |
| 2       |                                                  | <ul style="list-style-type: none"><li>Applied AI; Role of AI in Data Analysis; demo of Google Colab Notebook + AI model; Data collection; Survey data, open datasets, ethical collection practices</li></ul> | 2 hours  |
| 3       | Data Loading and Cleaning                        | <ul style="list-style-type: none"><li>Hands-on importing CSV/Excel/JSON; introduction to notebooks</li><li>Handling missing values, duplicates, formatting issues; outliers handing</li></ul>                | 2 hours  |
| 4       |                                                  | <ul style="list-style-type: none"><li>Data transformation, feature selection, automation tips using AI models</li></ul>                                                                                      | 2 hours  |
| 5       | Exploratory Data Analysis (EDA)                  | <ul style="list-style-type: none"><li>Descriptive statistics, mean, median, mode;</li><li>Visualizing distributions, correlation, relationships, scatterplots, bar/line charts</li></ul>                     | 2 hours  |
| 6       |                                                  | <ul style="list-style-type: none"><li>Advanced visualization: heatmaps, boxplots, interactive charts;</li><li>AI-assisted insights from the graph</li></ul>                                                  | 2 hours  |
| 7       | Applying AI to Qualitative and Quantitative Data | <ul style="list-style-type: none"><li>Simple predictive tasks, regression, classification, clustering with guidance from AI model</li><li>Results/ output description using AI model</li></ul>               | 2 hours  |
| 8       |                                                  | <ul style="list-style-type: none"><li>Text summarization, keyword extraction, sentiment analysis using prebuilt notebooks</li><li>Word cloud, topic modeling, Sankey plot</li></ul>                          | 2 hours  |
| 9       | Exam & Project                                   | Final exam and project demonstration.                                                                                                                                                                        | 4 hours  |